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Past Papers Advanced Higher Chemistry

2023 Marking Scheme

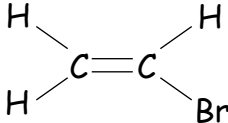
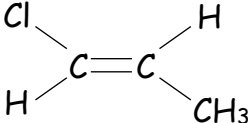
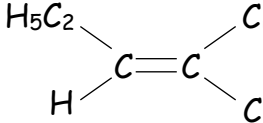
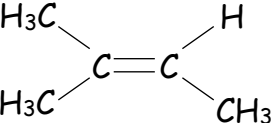
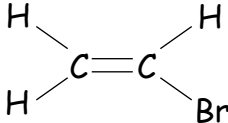
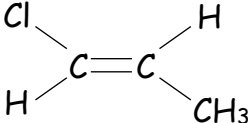
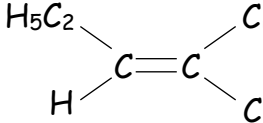
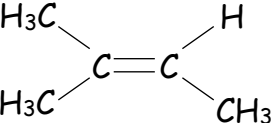
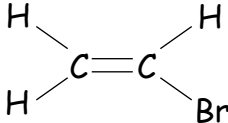
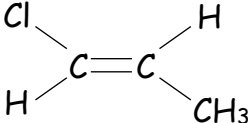
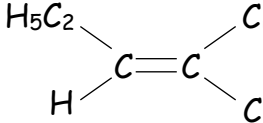
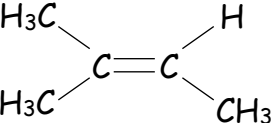
Grade Awarded	Mark Required		% candidates achieving grade
	(/120)	%	
A	77+	64.2%	27.2%
B	62+	51.7%	28.6%
C	48+	40.0%	22.8%
D	33+	27.5%	14.6%
No award	<33	<27.5%	6.7%

Section:	Multiple Choice	Extended Answer	Project
Average Mark:	15.7 /25	50.4 /95	No Project in 2023

2023 Adv Higher Chemistry Marking Scheme

MC Qu	Answer	% Correct	Reasoning																																																
1	D	89	☒ A Aufbau principle states that 4s fills before 3d shells ☒ B Aufbau principle states that 4s fills before 3d shells ☒ C 4s ¹ only occurs when 3d is either half-filled 3d ⁵ in Cr or completely full 3d ¹⁰ in Cu ☒ D 4s is filled before 3d																																																
2	C	68	<table><tr><th colspan="2">Linear</th><th colspan="2">Tetrahedral</th><th colspan="2">Trigonal Pyramidal</th><th colspan="2">Trigonal Bipyramidal</th></tr><tr><th colspan="2">2</th><th colspan="2">4</th><th colspan="2">4</th><th colspan="2">5</th></tr><tr><th colspan="2">electron pairs</th><th colspan="2">electron pairs</th><th colspan="2">electron pairs</th><th colspan="2">electron pairs</th></tr><tr><th>2</th><th>0</th><th>4</th><th>0</th><th>3</th><th>1</th><th>5</th><th>0</th></tr><tr><th>bonding pairs</th><th>lone pairs</th><th>bonding pairs</th><th>lone pairs</th><th>bonding pairs</th><th>lone pairs</th><th>bonding pairs</th><th>lone pairs</th></tr><tr><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td><td colspan="2"></td></tr></table>	Linear		Tetrahedral		Trigonal Pyramidal		Trigonal Bipyramidal		2		4		4		5		electron pairs		electron pairs		electron pairs		electron pairs		2	0	4	0	3	1	5	0	bonding pairs	lone pairs	bonding pairs	lone pairs	bonding pairs	lone pairs	bonding pairs	lone pairs								
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3	B	95	☒ A s-block found in groups 1+2 ☒ B p-block found in groups 3→0 ☒ C d-block found in transition metals between groups 2+3 ☒ D f-block found in bottom 2 rows of Periodic Table																																																
4	A	65	☒ A Heterogeneous catalysis work by adsorption of reactants onto active sites of catalyst and homogeneous catalyst are in same state as catalysts. ☒ B an incomplete d-subshell is required for the interaction of the catalyst with reactant(s) ☒ C heterogeneous catalyst is in a different state to the reactants ☒ D adsorption onto active sites is the mechanism of heterogeneous catalysts																																																
5	D	47	Halfway point of steep vertical region (neutralisation point) of line is approx. pH=6 ☒ A acidic pH at start of experiment ∴ alkali is being added to an acid ☒ B strong alkali/weak acid gives alkaline pH on neutralisation but pH is acidic at neutralisation ☒ C acidic pH at start of experiment ∴ alkali is being added to an acid ☒ D pH at neutralisation is acidic weak alkali and strong acid gives acidic pH on neutralisation																																																
6	D	69	<table><tr><th>Term</th><th>Bronsted-Lowry Definition</th></tr><tr><td>acid</td><td>Species which donates H⁺ and becomes conjugate base</td></tr><tr><td>base</td><td>Species which accepts H⁺ and becomes conjugate acid</td></tr><tr><td>conjugate acid</td><td>Species formed when base accepted an H⁺ ion</td></tr><tr><td>conjugate base</td><td>Species formed when acid donated an H⁺ ion</td></tr></table> H₃O⁺ ions being absorbed by conjugate base turning back into acid Weak acid donates H₃O⁺ ions and becomes conjugate base in the process.	Term	Bronsted-Lowry Definition	acid	Species which donates H ⁺ and becomes conjugate base	base	Species which accepts H ⁺ and becomes conjugate acid	conjugate acid	Species formed when base accepted an H ⁺ ion	conjugate base	Species formed when acid donated an H ⁺ ion																																						
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7	A	78	☒ A reactions which are feasible have negative values of ΔG and equilibrium favours products ☒ B a feasible reaction is one that tends towards the products rather than the reactants ☒ C a reaction with a positive value for standard free energy (ΔG) is not feasible ☒ D a reaction with a positive value for standard free energy (ΔG) is not feasible																																																
8	B	46	<table><tr><td>①</td><td>$\frac{1}{2}\text{N}_2 + \text{O}_2 \rightarrow \text{NO}_2$</td><td>$\Delta G^\circ = +51.8 \text{ kJ mol}^{-1}$</td></tr><tr><td>②</td><td>$\text{N}_2 + 2\text{O}_2 \rightarrow \text{N}_2\text{O}_4$</td><td>$\Delta G^\circ = +97.7 \text{ kJ mol}^{-1}$</td></tr><tr><td>① x -2</td><td>$2\text{NO}_2 \rightarrow \text{N}_2 + 2\text{O}_2$</td><td>$\Delta G^\circ = -103.6 \text{ kJ mol}^{-1}$</td></tr><tr><td>②</td><td>$\text{N}_2 + 2\text{O}_2 \rightarrow \text{N}_2\text{O}_4$</td><td>$\Delta G^\circ = +97.7 \text{ kJ mol}^{-1}$</td></tr><tr><td>cancel out</td><td>$2\text{NO}_2 \rightarrow \cancel{\text{N}_2} + \cancel{2\text{O}_2}$ $\cancel{\text{N}_2} + \cancel{2\text{O}_2} \rightarrow \text{N}_2\text{O}_4$</td><td>$\Delta G^\circ = -5.9 \text{ kJ mol}^{-1}$</td></tr><tr><td>Final equation</td><td>$2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$</td><td>$\Delta G^\circ = -5.9 \text{ kJ mol}^{-1}$</td></tr></table>	①	$\frac{1}{2}\text{N}_2 + \text{O}_2 \rightarrow \text{NO}_2$	$\Delta G^\circ = +51.8 \text{ kJ mol}^{-1}$	②	$\text{N}_2 + 2\text{O}_2 \rightarrow \text{N}_2\text{O}_4$	$\Delta G^\circ = +97.7 \text{ kJ mol}^{-1}$	① x -2	$2\text{NO}_2 \rightarrow \text{N}_2 + 2\text{O}_2$	$\Delta G^\circ = -103.6 \text{ kJ mol}^{-1}$	②	$\text{N}_2 + 2\text{O}_2 \rightarrow \text{N}_2\text{O}_4$	$\Delta G^\circ = +97.7 \text{ kJ mol}^{-1}$	cancel out	$2\text{NO}_2 \rightarrow \cancel{\text{N}_2} + \cancel{2\text{O}_2}$ $\cancel{\text{N}_2} + \cancel{2\text{O}_2} \rightarrow \text{N}_2\text{O}_4$	$\Delta G^\circ = -5.9 \text{ kJ mol}^{-1}$	Final equation	$2\text{NO}_2 \rightarrow \text{N}_2\text{O}_4$	$\Delta G^\circ = -5.9 \text{ kJ mol}^{-1}$																														
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9	C	55	Positive ΔS value means an increase in disorder. ☒ A burning solid carbon into the gas CO_2 is an increase in disorder and exothermic ☒ B water vapour turning into solid snowflakes is a decrease in disorder and exothermic ☒ C liquid ethoxyethane (ether) evaporating into gas is endothermic and increases disorder ☒ D two gases becoming one solid is a decrease in disorder and exothermic
10	A	76	Rate = $k[\text{X}][\text{Y}]$ means that both X and Y are first order and 1 molecule of each appears as reactants in the slow rate determining step. $\therefore \text{X} + \text{Y} \rightarrow \text{intermediate}$ The overall reaction is $\text{X} + 2\text{Y} \rightarrow \text{Z}$ so there must be a second step where the intermediate formed in (slow) rate determining step reacts with a further molecule of Y.
11	A	77	The single carbon - carbon bond in alkanes is a sigma bond. Sigma bonds are formed by the end-on overlap of atomic orbitals along the axis of the covalent bond. In a non-polar covalent bond in an alkane, the bonding molecular orbital is symmetrical about the midpoint between two atoms. (Polar covalent bonds result from bonding molecular orbitals that are asymmetric about the midpoint between two atoms.)
12	C	87	☒ A blue-green light is absorbed and not transmitted. Only red colour is transmitted. ☒ B electrons move from HOMO to LUMO to absorb a wavelength of light ☒ C electrons move from HOMO to LUMO and blue-green light absorbed and red transmitted ☒ D electrons move from HOMO to LUMO to absorb a wavelength of light
13	D	87	☒ A ethanol contains O-H bonds and has hydrogen bonding between molecules ☒ B ethylamine contains N-H bonds and has hydrogen bonding between molecules ☒ C ethanoic acid contains O-H bonds and has hydrogen bonding between molecules ☒ D ethoxyethane has no N-H, O-H or H-F bonds and no hydrogen bonding in pure substance
14	A	70	secondary bromoalkane secondary alcohol Ketone
15	C	71	2-methylbutanal 2-methylbutan-1-ol $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CHO}$ is an aldehyde called 2-methylbutanal LiAlH_4 is the reducing agent lithium aluminium hydride - Aldehydes are reduced to primary alcohols by LiAlH_4
16	B	63	$\text{C}_4\text{H}_8\text{O}_2 + \text{NH}_3 \rightarrow \text{X} \rightarrow \text{Y} + \text{H}_2\text{O}$ $\text{C}_4\text{H}_8\text{O}_2 + \text{NH}_3 \rightarrow \text{C}_4\text{H}_{11}\text{NO}_2 \rightarrow \text{C}_4\text{H}_9\text{NO} + \text{H}_2\text{O}$ ☒ A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ has a molecular formula of $\text{C}_4\text{H}_7\text{N}$ ☒ B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CONH}_2$ has a molecular formula of $\text{C}_4\text{H}_9\text{NO}$ ☒ C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_2$ has a molecular formula of $\text{C}_4\text{H}_{11}\text{N}$ ☒ D $\text{CH}_3\text{CH}_2\text{CH}_2\text{COONH}_2$ has a molecular formula of $\text{C}_4\text{H}_9\text{NO}_2$

17	A	87	Methoxypropane has the structure: $\begin{array}{ccccccc} & & & & \text{H} & & \text{H} & \text{H} & \text{H} \\ & & & & & & & & \\ \text{H} & - & \text{C} & - & \text{O} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & & & & & & & & \\ & & & & \text{H} & & \text{H} & \text{H} & \text{H} \end{array}$ It has a molecular formula of $\text{C}_4\text{H}_{10}\text{O}$ ☒ A butanone $\text{CH}_3\text{CH}_2\text{COCH}_3$ has a molecular formula of $\text{C}_4\text{H}_8\text{O}$ ☒ B ethoxyethane $\text{CH}_3\text{CH}_2\text{OCH}_2\text{CH}_3$ has a molecular formula of $\text{C}_4\text{H}_{10}\text{O}$ ☒ C butan-2-ol $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{CH}_3$ has a molecular formula of $\text{C}_4\text{H}_{10}\text{O}$ ☒ D 2-methylpropan-1-ol $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{OH}$ has a molecular formula of $\text{C}_4\text{H}_{10}\text{O}$																				
18	B	49	<table><tr><td>A</td><td>B</td><td>C</td><td>D</td></tr><tr><td>CH_2CHBr</td><td>CHClCHCH_3</td><td>$\text{CH}_3\text{CH}_2\text{CHCl}_2$</td><td>$\text{CH}_3\text{C}(\text{CH}_3)\text{CHCH}_3$</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td>bromoethane</td><td>trans-1-chloropropene</td><td>1,1-dichlorobut-1-ene</td><td>2-methylbut-2-ene</td></tr><tr><td>Left end of $\text{C}=\text{C}$ double bond has two H groups attached. This prevents it from being a geometric isomer</td><td>There are two (non-identical) groups, one on each end of the $\text{C}=\text{C}$ double bond. This makes it a geometric isomer</td><td>Left end of $\text{C}=\text{C}$ double bond has two H groups attached. This prevents it from being a geometric isomer</td><td>Left end of $\text{C}=\text{C}$ double bond has two CH_3 groups attached. This prevents it from being a geometric isomer</td></tr></table>	A	B	C	D	CH_2CHBr	CHClCHCH_3	$\text{CH}_3\text{CH}_2\text{CHCl}_2$	$\text{CH}_3\text{C}(\text{CH}_3)\text{CHCH}_3$					bromoethane	trans-1-chloropropene	1,1-dichlorobut-1-ene	2-methylbut-2-ene	Left end of $\text{C}=\text{C}$ double bond has two H groups attached. This prevents it from being a geometric isomer	There are two (non-identical) groups, one on each end of the $\text{C}=\text{C}$ double bond. This makes it a geometric isomer	Left end of $\text{C}=\text{C}$ double bond has two H groups attached. This prevents it from being a geometric isomer	Left end of $\text{C}=\text{C}$ double bond has two CH_3 groups attached. This prevents it from being a geometric isomer
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19	A	47	☒ A in mass spectroscopy, the molecule breaks up during the process to give different lines ☒ B in infrared spectroscopy, molecule has certain bonds stretched at certain wavenumbers ☒ C in melting point analysis, a pure substance will melt first but will freeze back again. ☒ D Proton NMR uses radio waves to flip nuclei but does not destroy the sample.																				
20	D	65	<table><tr><td>Type of Drug</td><td>Binding Site</td><td>Response</td></tr><tr><td>Agonist</td><td>receptor binding site</td><td>produces a response similar to the body's natural response</td></tr><tr><td>Antagonist</td><td>receptor binding site</td><td>blocks the body's natural response</td></tr><tr><td>Enzyme Inhibitor</td><td>enzyme active site</td><td>blocks the action of the enzyme</td></tr></table>	Type of Drug	Binding Site	Response	Agonist	receptor binding site	produces a response similar to the body's natural response	Antagonist	receptor binding site	blocks the body's natural response	Enzyme Inhibitor	enzyme active site	blocks the action of the enzyme								
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21	D	65	☒ A As doxepin is only molecule with O atom at top, this part not in common fragment ☒ B Molecule B is incorrect at bottom the nitrogen part is rotatory and will flip round ☒ C Doxepin lacks the -Cl group on top left of molecule $\therefore$ Cl not in common fragment ☒ D Molecule D is correct as the nitrogen group is rotary & flips round and found in all drugs																				
22	B	87	<table><tr><td>Element</td><td>V</td><td>O</td></tr><tr><td>Mass or %</td><td>61.4%</td><td>38.6</td></tr><tr><td>No. of moles (divide % by gfm)</td><td>$\frac{61.4}{50.9}$ = 1.21</td><td>$\frac{38.6}{16}$ = 2.41</td></tr><tr><td>Mole ratio (divide through by smallest value)</td><td>$\frac{1.21}{1.21}$ = 1</td><td>$\frac{2.41}{1.21}$ = 1.99 = 2</td></tr><tr><td>Empirical Formula</td><td colspan="2">VO_2</td></tr></table>	Element	V	O	Mass or %	61.4%	38.6	No. of moles (divide % by gfm)	$\frac{61.4}{50.9}$ = 1.21	$\frac{38.6}{16}$ = 2.41	Mole ratio (divide through by smallest value)	$\frac{1.21}{1.21}$ = 1	$\frac{2.41}{1.21}$ = 1.99 = 2	Empirical Formula	VO_2						
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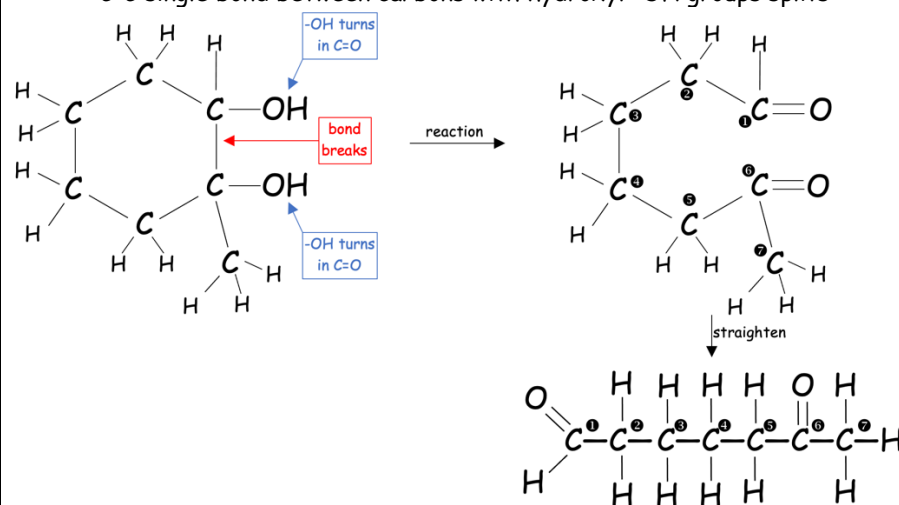
23	C	49	A primary standard must:									
			<table><tr><td>be available in a high state of purity</td><td>be stable when solid and in solution</td><td>be soluble</td><td>have a reasonably high GFM</td></tr></table> <table><tr><td>Examples of primary standards include:</td><td>sodium carbonate Na₂CO₃</td><td>hydrated oxalic acid H₂C₂O₄·2H₂O</td><td>potassium hydrogen phthalate KH(C₈H₄O₄)</td></tr><tr><td></td><td>silver nitrate AgNO₃</td><td>potassium iodate KIO₃</td><td>potassium dichromate K₂Cr₂O₇</td></tr></table> ☒ A calcium carbonate is insoluble and therefore not a primary standard ☒ B Hydrochloric acid ☒ C sodium carbonate meets the four criteria listed in table above ☒ D Sodium hydroxide is not a primary standard as it has a relatively low GFM, is unstable as a solid (absorbs moisture) and unstable as a solution. Sodium hydroxide solution must be standardised before being used in volumetric analysis.	be available in a high state of purity	be stable when solid and in solution	be soluble	have a reasonably high GFM	Examples of primary standards include:	sodium carbonate Na ₂ CO ₃	hydrated oxalic acid H ₂ C ₂ O ₄ ·2H ₂ O	potassium hydrogen phthalate KH(C ₈ H ₄ O ₄)	
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	silver nitrate AgNO ₃	potassium iodate KIO ₃	potassium dichromate K ₂ Cr ₂ O ₇									
24	B	61	420 ppm = 420mg CO₂ per 1kg atmosphere 1kg atmosphere = 0.42g CO₂ 5.1×10¹⁸kg atmosphere = 0.42g CO₂ × 5.1×10¹⁸ = 2.142×10¹⁸g CO₂ = 2.1×10¹⁵kg CO₂									
25	B	41	Pb²⁺ + CrO₄²⁻ → PbCrO₄ 1mol 1mol PbCrO₄ + 2Cl⁻ → PbCl₂ + CrO₄²⁻ 1mol 1mol 2CrO₄²⁻ + 6I⁻ + 16H⁺ → 2Cr³⁺ + 3I₂ + 8H₂O 2mol 3mol 1mol 1.5mol									

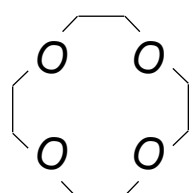
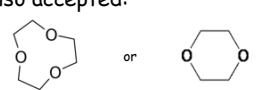
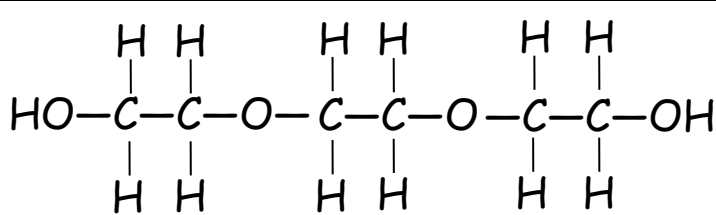
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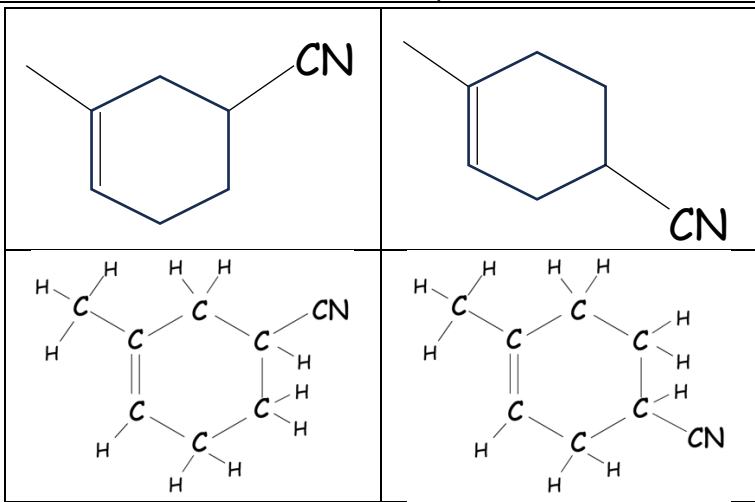
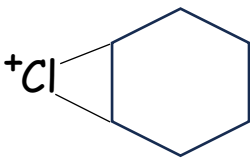
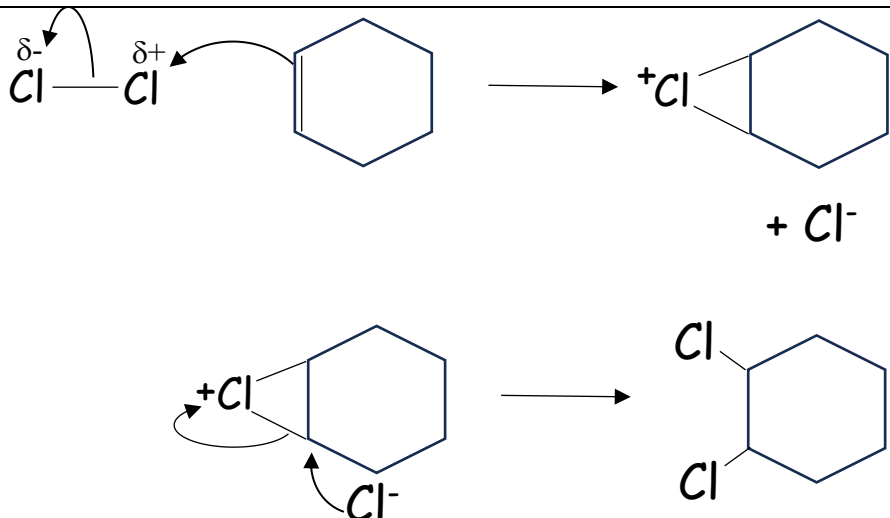
Long Qu	Answer	Reasoning					
1a(i)	Answer to include:	1 st mark:	(Excited) electrons fall to a lower energy level				
		2 nd mark:	Difference in energy levels corresponds to $\left[\begin{array}{c} \text{red light} \\ \text{light} \\ \text{energy} \end{array} \right]$ emitted.				
1a(ii)		Li has electronic configuration of $1s^2 2s^1$. Li^+ has electronic configuration of $1s^2$					
		n	1s subshell is in 1 st shell and has a value for Principal quantum number (n) of n=1				
		l	s subshells have an angular momentum quantum number (l) of l =0				
	1	m_l	s subshells have a magnetic quantum number (m_l) of -l to +l $\therefore$ 0				
		m_s	All electrons either have a spin magnetic quantum number of either $+\frac{1}{2}$ or $-\frac{1}{2}$				
1a(iii)A	+7 or VII	Oxidation state of O = -2 $\therefore$ 4xO in ClO_4^- combine to equal -8. Perchlorate ClO_4^- has charge of -1 over whole ion $\therefore$ the oxidation state of Cl must be +7					
1a(iii)B	9	Bond	Number of bonds	Number of sigma	Number of Pi bonds		
		N - H	3	$3 \times 1 = 3$	0		
		N - C	1	1	0		
		C = N	1	1	1		
		C - N	1	1	0		
		N - N	2	$2 \times 1 = 2$	0		
		N = N	1	1	1		
		Total			9	2	
		1b(i)A	-1648	ΔH°	=	$\Sigma \Delta H_f^\circ(\text{products})$	-
	=			(2x-824)	-	(4x0) + (3x0)	
	=			-1648	-	0	
	=			-1648 kJ mol ⁻¹			
1b(i)B	-549.4			ΔS°	=	$\Sigma S^\circ(\text{products})$	-
			=	(2x87.4)	-	(4x27.3) + (3 x205)	
			=	174.8	-	(109.2 + 615)	
			=	174.8	-	724.2	
			=	-549.4 J K ⁻¹ mol ⁻¹			
1b(i)C	3000	$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = 0 \therefore T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{-1648 \times 1000 \text{ J mol}^{-1}}{-549.4 \text{ J K}^{-1} \text{ mol}^{-1}} = 3000\text{K}$					
1b(ii)	2694g	gfm $NaClO_3 = (1 \times 23) + (1 \times 35.5) + (3 \times 16) = 23 + 35.5 + 48 = 106.5\text{g}$					
		1 person in 1 minute	uses	0.380	litres oxygen		
		5 persons in 1 minute	uses	0.380	litres oxygen x 5		
					= 1.9	litres oxygen	
		5 persons in 1 hour	uses	1.9	litres oxygen x 60		
					= 114	litres oxygen	
		5 persons in 8 hours	uses	114	litres oxygen x 8		
					= 912	litres oxygen	
36litres oxygen	produced from	1mol $NaClO_3$					
	912litres oxygen	produced from	1mol $NaClO_3 \times \frac{912}{36}$				
			= 25.3mol $NaClO_3$				
mass = no. of mol x gfm = 25.3 x 106.5 = 2694g							
2a(i)	1 st Order	Experiments	$[HgCl_2]$	$[C_2O_4^{2-}]$	Effect of Rate	Order	
		3+2	x2	same	x2	$[HgCl_2]^1$	
2a(ii)	2 nd Order	Experiments	$[HgCl_2]$	$[C_2O_4^{2-}]$	Effect of Rate	Order	
		1+2	same	x2	x4	$[C_2O_4^{2-}]^2$	

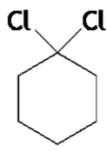
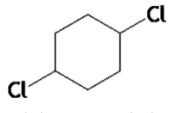
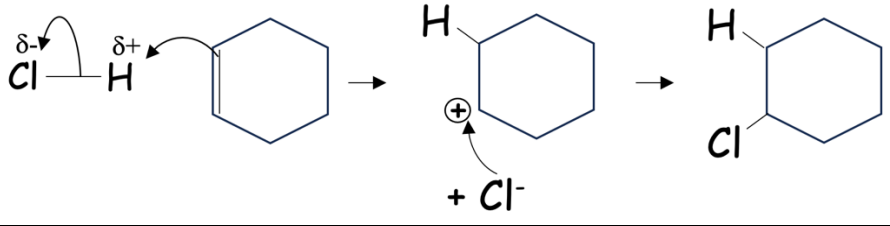
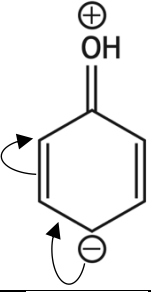
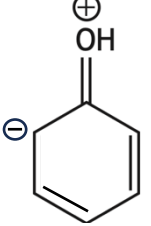
2b	rate = $k[\text{HgCl}_2][\text{C}_2\text{O}_4^{2-}]^2$	rate = $k \times [\text{HgCl}_2]^1 \times [\text{C}_2\text{O}_4^{2-}]^2 = k[\text{HgCl}_2][\text{C}_2\text{O}_4^{2-}]^2$																		
2c(i)	$2.56 \times 10^{-4} \text{ l}^2 \text{ mol}^{-2} \text{ s}^{-1}$ Or $0.000256 \text{ l}^2 \text{ mol}^{-2} \text{ s}^{-1}$	Using Experiment 1 (although experiments 2+3 can also be used) $k = \frac{\text{rate}}{[\text{HgCl}_2][\text{C}_2\text{O}_4^{2-}]^2} = \frac{0.860 \times 10^{-6} \text{ mol l}^{-1} \text{ s}^{-1}}{0.0840 \text{ mol l}^{-1} \times (0.200)^2 \text{ mol}^2 \text{ l}^{-2}} = 2.56 \times 10^{-4} \text{ l}^2 \text{ mol}^{-2} \text{ s}^{-1}$																		
2c(ii)	0.508	For experiment 4: $\text{Rate} = k \times [\text{HgCl}_2] \times [\text{C}_2\text{O}_4^{2-}]^2$ $2.11 \times 10^{-6} = 2.56 \times 10^{-4} \times 0.0320 \times [\text{C}_2\text{O}_4^{2-}]^2$ $[\text{C}_2\text{O}_4^{2-}]^2 = \frac{2.11 \times 10^{-6}}{2.56 \times 10^{-4} \times 0.0320} = 0.258 \text{ mol}^2 \text{ l}^{-2}$ $[\text{C}_2\text{O}_4^{2-}] = \sqrt{0.258} \text{ mol l}^{-1} = 0.508 \text{ mol l}^{-1}$																		
3a(i)	Add more/excess calcium choride solution (to see if further precipitation occurs)	When precipitation appears to be complete, the mixture is allowed to settle and more reactant is added to see if there is any further precipitation/cloudiness showing the reaction is incomplete and further reactant is added. If there is no more precipitation/cloudiness then the precipitate can be removed by vacuum filtration.																		
3a(ii)	The precipitate should be dried/heated in an oven and weighed until constant mass	When precipitate is removed from the filtrate by vacuum filtration, some of the water remains in the precipitate residue. Heating the precipitate in an oven will drive off water. When the precipitate is removed from the oven it is cooled in a desiccator and the mass measured in a balance. Heating in oven, cooling in a desiccator and weight on balance is repeated until constant mass is achieved.																		
3b	0.59	$\text{gfm Ca(COO)}_2 = (1 \times 40.1) + (2 \times 12) + (4 \times 16) = 40.1 + 24 + 64 = 128.1\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.075}{128.1} = 5.85 \times 10^{-4} \text{mol}$ $\begin{array}{ccc} (\text{COOH})_2 + \text{CaCl}_2 & \longrightarrow & \text{Ca(COO)}_2 + 2\text{HCl} \\ 1\text{mol} & & 1\text{mol} \\ 5.85 \times 10^{-4} \text{mol} & & 5.85 \times 10^{-4} \text{mol} \end{array}$ $\text{gfm (COOH)}_2 = (2 \times 12) + (4 \times 16) + (2 \times 1) = 24 + 64 + 2 = 90\text{g}$ $\text{mass} = \text{no. of mol} \times \text{gfm} = 5.85 \times 10^{-4} \times 90 = 0.0527\text{g}$ $\% \text{ mass} = \frac{\text{mass of oxalic acid}}{\text{mass of leaves}} \times 100 = \frac{0.0527}{8.975} \times 100 = 0.59\%$																		
3c	One answer from:	<table><tr><td>Not all oxalic acid extracted out</td><td>Not soaked in water for long enough</td><td>Not all precipitated out/not gone to completion</td></tr><tr><td>Not enough CaCl_2 added</td><td>Transfer loss</td><td>Natural variation in samples</td></tr></table>	Not all oxalic acid extracted out	Not soaked in water for long enough	Not all precipitated out/not gone to completion	Not enough CaCl_2 added	Transfer loss	Natural variation in samples												
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5a	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$	Charge in Nickel ion = +2 as H_2O is a neutral ligand so +2 charge on complex comes from Ni^{2+} ion. Ni atom: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$ 4s electrons are removed before 3d electrons Ni^{2+} ion: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$																		

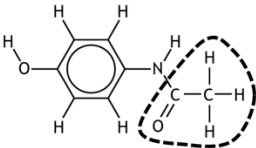
5b	Hexaamminenickel(II)	Hexaamminenickel(II) = [Ni(NH ₃) ₆] ²⁺					
		no. of ligands	NH ₃ ligand	metal name	charge on metal ion		
		Neutral ligands include:		Negative Ligands include:	Central Ion:	Charge:	
		Ligand	Name	Ligand	Name	Positive Complex: metals keep their name	Charge of central ion is converted into roman numerals and put in brackets
		H ₂ O	aqua	Chloride Cl ⁻	chlorido	Negative Complex: Metals end in ATE e.g. Cuprate, Ferrate, Cobaltate	
		NH ₃	ammine	Cyanide CN ⁻	cyanido		
		CO	carbonyl	Nitrite NO ₂ ⁻	nitrito		
5c(i)	Bidentate	A bidentate ligand has two parts of the ligand that can donate pairs of electrons to the central metal ion at the centre of the complex.					
5c(ii)	6	6 different pairs of electrons across three bidentate ligands are donated to central metal ion. This gives a co-ordination number of 6.					
5d(i)	Answer to include:	1 st mark:	Electrons move to higher energy d orbitals				
		2 nd mark:	purple light 380-400nm light red and blue light complementary colour (to green)				} absorbed
5d(ii)	One answer from:	Ammonia absorbs { shorter wavelength higher energy higher frequency uv					
6a(i)	$K = \frac{[HI]^2}{[H_2][I_2]}$	For reaction: $aA + bB \rightleftharpoons cC + dD$ $K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$					
6a(ii)	981.8	I_2 at start = 0.25mol I_2 at equilibrium = 0.015mol $\therefore I_2$ reacted = 0.25mol - 0.015mol = 0.235mol $H_2 + I_2 \rightleftharpoons 2HI$ 1mol 1mol 2mol 0.235mol 0.235mol 0.470mol (reacted) (reacted) (formed) H_2 remaining at equilibrium = 0.25mol - 0.235mol = 0.015mol $K = \frac{[HI]^2}{[H_2][I_2]} = \frac{(0.470)^2}{0.015 \times 0.015} = 981.8$					
6a(iii)	207.2	$E = \frac{L \times h \times c}{\lambda} = \frac{6.02 \times 10^{23} \text{ mol}^{-1} \times 6.63 \times 10^{-34} \text{ J s} \times 3 \times 10^8 \text{ m s}^{-1}}{578 \times 10^{-9} \text{ m}}$ $= 207159 \text{ J mol}^{-1}$ $= 207.2 \text{ kJ mol}^{-1}$					
6b(i)	Buchner Funnel or Hirsch Funnel or Sintered Glass Funnel	The Buchner funnel allows vacuum filtration to place which speeds up the flow of filtrate through the Buchner funnel. The reduced pressure provides a faster means to separate precipitate from filtrate					

6b(ii)	87.4%	$\text{gfm } I_2 = (2 \times 126.9) = 253.8\text{g}$ $\text{no. of mol } I_2 = \frac{\text{mass}}{\text{gfm}} = \frac{285}{253.8} = 1.12\text{mol}$ $I_2 + H_2S \longrightarrow 2HI + S$ 1mol 1.12mol 2mol $2.25\text{mol (theoretical)}$ $\text{gfm HI} = (1 \times 1) + (1 \times 126.9) = 1 + 126.9 = 127.9\text{g}$ $\text{no. of mol HI} = \frac{\text{mass}}{\text{gfm}} = \frac{251}{127.9} = 1.96 \text{ mol (actual)}$ $\% \text{ Yield} = \frac{\text{actual}}{\text{theoretical}} \times 100 = \frac{1.96}{2.25} \times 100 = 87.4\%$												
6c		Notes on example: - Hydroxyl -OH groups become carbonyl C=O groups - C-C single bond between carbons with hydroxyl -OH groups splits 												
7a	Concentration of dissolved CO_2 decreases as equilibrium moves to left.	Increase in temperature favours the endothermic reaction Reverse reaction is favoured as reverse reaction is endothermic Reverse reaction decreases $H_2CO_{3(aq)}$ concentration and releases $CO_{2(g)}$ into air.												
7b(i)	Carbonic acid is a weaker acid as it has higher pK_a / lower K_a or Ethanoic acid is a stronger acid as it has lower pK_a / higher K_a	<table><tr><th>Weak Acid</th><th>K_a</th><th>pK_a</th></tr><tr><td>Carbonic acid</td><td>4.5×10^{-7}</td><td>6.35</td></tr><tr><td>Ethanoic acid</td><td>1.7×10^{-5}</td><td>4.76</td></tr><tr><td>Conclusion</td><td>Carbonic acid is a lower strength acid/weaker acid as it has a smaller value of K_a</td><td>Carbonic acid is a lower strength acid/weaker acid as it has a higher value of pK_a</td></tr></table>	Weak Acid	K_a	pK_a	Carbonic acid	4.5×10^{-7}	6.35	Ethanoic acid	1.7×10^{-5}	4.76	Conclusion	Carbonic acid is a lower strength acid/weaker acid as it has a smaller value of K_a	Carbonic acid is a lower strength acid/weaker acid as it has a higher value of pK_a
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7b(ii)	Base (or proton acceptor)	<table><tr><td>$H_2CO_{3(aq)}$ Acid loses a proton/H^+ (proton donor)</td><td>$H_2O(l)$ Base gains a proton/H^+ (proton acceptor)</td><td>$H_3O^+(aq)$ Conjugate Acid formed when base accepts a proton/H^+</td><td>$HCO_3^-(aq)$ Conjugate base formed when acid loses a proton/H^+</td></tr></table>	$H_2CO_{3(aq)}$ Acid loses a proton/ H^+ (proton donor)	$H_2O(l)$ Base gains a proton/ H^+ (proton acceptor)	$H_3O^+(aq)$ Conjugate Acid formed when base accepts a proton/ H^+	$HCO_3^-(aq)$ Conjugate base formed when acid loses a proton/ H^+								
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7c(i)	6.31×10^{-9}	$pH = 8.2$ $-\log_{10}[H_3O^+] = 8.2$ $\log_{10}[H_3O^+] = -8.2$ $[H_3O^+] = 10^{-8.2}$ $[H_3O^+] = 6.31 \times 10^{-9} \text{ mol l}^{-1}$												

7c(ii)	99.7%	$\begin{aligned} \text{pH} &= 7.9 \\ -\log_{10}[\text{H}_3\text{O}^+] &= 7.9 \\ \log_{10}[\text{H}_3\text{O}^+] &= -7.9 \\ [\text{H}_3\text{O}^+] &= 10^{-7.9} \\ [\text{H}_3\text{O}^+] &= 1.26 \times 10^{-8} \text{ mol l}^{-1} \end{aligned}$		$\% \text{ Increase} = \frac{\text{difference in value}}{\text{original value}} \times 100$ $\% \text{ Increase} = \frac{1.26 \times 10^{-8} - 6.31 \times 10^{-9}}{6.31 \times 10^{-9}} \times 100$ $\% \text{ Increase} = \frac{6.29 \times 10^{-9}}{6.31 \times 10^{-9}} \times 100$ $\% \text{ Increase} = 99.7\%$																	
7d(i)	Answer to include:	Reacting known mass of seashells with an excess volume of a hydrochloric acid of known concentration	The resulting mixture is then titrated with sodium hydroxide to work out the number of moles of hydrochloric acid in excess	From the initial number of moles of hydrochloric acid, the number of moles used in the reaction can be determined.	The initial number of moles of the calcium carbonate in seashells being analysed can then be calculated.																
7d(i)	Answer to include: All 4 points = 2marks 2 or 3 points = 1 mark	React calcium carbonate /shells with HCl/acid	Known quantity/moles/volume and concentration of HCl/acid	idea of excess HCl/acid	titrate with NaOH.																
7d(ii)	Repeat using pure calcium carbonate	In a control experiment, a known mass of pure calcium carbonate can assess the accuracy of the reaction method.																			
8a(i)	15-crown-5	In example given 18-crown-6: 18 refers to the number of atoms in the ring and 6 refers to number of oxygens The question: 15 atoms in the ring and 5 oxygen atoms in ring $\therefore$ 15-crown-5																			
8a(ii)	 Also accepted: 	<table><tr><th>Element</th><th>Member of Group 1</th><th>Number of O in ether</th><th>Name</th></tr><tr><td>Li</td><td>1st</td><td>4</td><td>12-crown-4</td></tr><tr><td>Na</td><td>2nd</td><td>5</td><td>15-crown-5</td></tr><tr><td>K</td><td>3rd</td><td>6</td><td>18-crown-6</td></tr></table>				Element	Member of Group 1	Number of O in ether	Name	Li	1 st	4	12-crown-4	Na	2 nd	5	15-crown-5	K	3 rd	6	18-crown-6
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8a(iii)	Can donate lone (non-bonding) pairs of electrons	Ligands have lone pairs of electrons which form dative covalent bonds by donating pairs of electrons to the central metal ion forming dative covalent bonds. In crown ethers, the lone pair electrons are pointing into the centre of the ring where the metal ion is located.																			
8b(i)	$\text{C}_6\text{H}_{14}\text{O}_4$																				
8b(ii)	Condensation or nucleophilic substitution	H from one molecule and OH from next molecule are removed as the molecules join together forming HCl																			
8c	2.43×10^{-3} or 0.0024	$\text{gfm } 18\text{-Crown-6} = 264\text{g}$ $\text{no. of mol} = \frac{\text{mass}}{\text{gfm}} = \frac{0.225}{264} = 8.52 \times 10^{-4} \text{ mol}$ As K ⁺ ions and 18-crown-6 are in 1:1 ratio: $8.52 \times 10^{-4} \text{ mol K}^+$ ions $\text{concentration} = \frac{\text{no. of mol}}{\text{volume}} = \frac{8.52 \times 10^{-4} \text{ mol}}{0.350 \text{ litres}} = 2.43 \times 10^{-3} \text{ mol l}^{-1}$																			

9	Open Ended Question:	3 mark answer	2 mark answer	1 mark answer												
	<table border="1"> <tr> <th colspan="4">Candidate Marks</th> </tr> <tr> <th>0 marks</th> <th>1 mark</th> <th>2 marks</th> <th>3 marks</th> </tr> <tr> <td>41%</td> <td>43%</td> <td>13%</td> <td>3%</td> </tr> </table>	Candidate Marks				0 marks	1 mark	2 marks	3 marks	41%	43%	13%	3%	Demonstrates a <u>good understanding</u> of the chemistry involved. A good comprehension of the chemistry has provided in a logically correct, including a statement of the principles involved and the application of these to respond to the problem.	Demonstrates a <u>reasonable understanding</u> of the chemistry involved, making some statement(s) which are relevant to the situation, showing that the problem is understood.	Demonstrates a <u>limited understanding</u> of the chemistry involved. The candidate has made some statement(s) which are relevant to the situation, showing that at least a little of the chemistry within the problem is understood.
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10a(i)	One answer from:	Adjacent (unhybridised) p orbitals that overlap/form a molecular orbital across a system/number of carbon atoms	Electrons delocalised across a number of carbon atoms	Alternating single and double bonds												
10a(ii)	sp^2	C=C double bonds contain sp^2 hybridisation with a C-C sigma bond down the axis of the bond and side on overlap of unhybridised orbitals to form the pi bond.														
10a(iii)	One from:															
10b(i)	Repulsion from pi electrons or double bond	The electrons Cl-Cl bond are repelled by the pi electrons in the C=C double bond and this polarises the Cl-Cl bond forming a δ^+ end and a δ^- end														
10b(ii)																
10b(iii)A	Restricted rotation/ lack of free rotation (around the single bonds) in the ring	The ring of six carbons prevents the rotation so there are two locked positions for side groups above and below the ring with no rotation. - if both side groups are above ring then this is the cis geometric isomer - if one side group is above ring and the other side group below ring then this is the trans geometric isomer. A C=C double bond also has a lack of rotation in the bond due to the Pi bond so that also has the geometric isomers														

10b(iii)B	structure drawn of 1,1-dichlorocyclohexane  or  1,4-dichlorocyclohexane	An optical isomer has a chiral carbon with four different groups attached. 1,1-dichlorocyclohexane has no carbons with four different groups. C_1 has two chlorine atoms and the other carbons each have two hydrogens. 1,4-dichlorocyclohexane lacks a chiral carbons as C_1 and C_4 each have a hydrogen atom and a chlorine atom but the ring of carbons is symmetrical and are identical regardless of which way you go around the ring.									
10c(i)	Diagram showing: 1 mark for curly arrows 1 mark for correct intermediate										
10c(ii)	1.08	$e = 1.60 \times 10^{-19} C$ partial charge = 0.178 $Q = \text{partial charge} \times e = 0.178 \times 1.60 \times 10^{-19} C = 2.848 \times 10^{-20} C$ $Q = 2.848 \times 10^{-20} C$ $r = 0.127 \text{ nm} = 0.127 \times 10^{-9} \text{ m} = 1.27 \times 10^{-10} \text{ m}$ $\mu = Q \times r = 2.848 \times 10^{-20} C \times 1.27 \times 10^{-10} \text{ m} = 3.62 \times 10^{-30} \text{ Cm}$ $3.34 \times 10^{-30} \text{ Cm} = 1 \text{ D}$ $3.62 \times 10^{-30} \text{ Cm} = 1 \text{ D} \times \frac{3.62 \times 10^{-30} \text{ Cm}}{3.34 \times 10^{-30} \text{ Cm}}$ $= 1.08 \text{ D}$									
11a(i)	NO_2^+	Nitration electrophilic substitution reactions involve concentrated nitric acid and concentrated sulphuric acid and forms the electrophile nitronium NO_2^+ ion $\text{HNO}_3 + \text{H}_2\text{SO}_4 \longrightarrow \text{NO}_2^+ + \text{H}_3\text{O}^+ + \text{HSO}_4^-$ concentrated concentrated nitronium hydronium hydrogensulphate nitric acid sulphuric acid ion ion ion									
11a(ii)A											
11a(ii)B											
11b	Reduction	<table border="1"> <thead> <tr> <th colspan="3">Decrease in the oxygen : hydrogen ratio</th> </tr> <tr> <th>Formula</th> <th>$\text{C}_6\text{H}_4\text{O}_3$</th> <th>$\text{C}_6\text{H}_7\text{O}$</th> </tr> </thead> <tbody> <tr> <td>O : H ratio</td> <td>3 : 4 1 : 1.33</td> <td>1 : 7</td> </tr> </tbody> </table>	Decrease in the oxygen : hydrogen ratio			Formula	$\text{C}_6\text{H}_4\text{O}_3$	$\text{C}_6\text{H}_7\text{O}$	O : H ratio	3 : 4 1 : 1.33	1 : 7
Decrease in the oxygen : hydrogen ratio											
Formula	$\text{C}_6\text{H}_4\text{O}_3$	$\text{C}_6\text{H}_7\text{O}$									
O : H ratio	3 : 4 1 : 1.33	1 : 7									

11c	Ethanoic acid or Ethanoic anhydride or Ethanoyl chloride	Amines react with carboxylic acids to form alkylammonium salts and form an amide when heated.							
11d	Answer to include:	2 marks for all 4 points in table. 1 mark for 2 or 3 points in table							
		minimum/small volume (of solvent)	hot water/solvent	cool	filter				
11e(i)A	C=O or Carbonyl								
11e(i)B	Overlap of peaks from different bonds with one example as shown:	O-H peak in same region as $\begin{Bmatrix} \text{N-H} \\ \text{C-H} \end{Bmatrix}$ or N-H peak in same region as $\begin{Bmatrix} \text{O-H} \\ \text{C-H} \end{Bmatrix}$							
11e(ii)A	CH ₃ -C=O circled: 	Chemical Shift (ppm)	2.1	6.7	7.3	9.1	9.8		
		Height of Curve (mm)	50	33	33	17	17		
		No of Hydrogens in Group	3	2	2	1	1		
		Multiplet	singlet	doublet	doublet	singlet	singlet		
		No of Hydrogen next door	0	2		0	0		
11e(ii)B	<table border="1"><tr><td>Doublet</td></tr><tr><td>Doublet</td></tr></table>	Doublet	Doublet	Group	CH ₃ C=O	ArH?		-HN-C=O	Ar-OH
Doublet									
Doublet									
11e(iii)A	Sample is pure with no extra peaks or only 5 (hydrogen) peaks	The same number of peaks with the same chemical shift indicated the sample is pure as contaminants would add other peaks to the NMR spectrum as they would containing other chemical groups.							
11e(iii)B	Mixed melting point or TLC or Chromatography	The melting point of the recrystallised paracetamol is measured using melting point apparatus. Pure paracetamol is added to see if the melting point is the same. If the melting point alters then the sample is not pure.							

2023 Advanced Higher Chemistry Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	0	7	4	89	0	0
2	C	9	6	68	17	1	0
3	B	1	95	3	0	0	0
4	A	65	1	3	31	0	0
5	D	4	29	19	47	0	0
6	D	11	8	11	69	0	0
7	A	78	10	10	2	0	0
8	B	9	46	7	37	1	0
9	C	10	16	55	18	1	0
10	A	76	4	15	5	0	0
11	A	77	16	5	1	0	0
12	C	1	1	87	11	0	0
13	D	1	10	1	87	0	0
14	A	70	10	8	11	1	0
15	C	8	11	71	10	1	0
16	B	17	63	16	4	1	0
17	A	87	4	4	4	1	0
18	B	11	49	17	22	1	0
19	A	47	9	27	17	1	0
20	D	6	26	4	65	0	0
21	D	6	26	3	65	0	0
22	B	2	87	8	3	0	0
23	C	28	11	49	11	1	0
24	B	6	61	12	21	0	0
25	B	6	41	43	9	1	0

2023 Advanced Higher Chemistry Section 2 Marks Data Analysis

2023 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1ai	2	0.9	46.3%	38	0.8%
Q1aii	1	0.5	51.3%	19	0.1%
Q1aiiiA	1	0.8	82.0%	44	2.7%
Q1aiiiB	1	0.8	84.7%	37	0.3%
Q1biA	1	0.7	65.5%	37	1.8%
Q1biB	1	0.7	69.5%	37	1.3%
Q1biC	2	1.6	77.9%	55	3.5%
Q1bii	2	1.4	69.6%	37	1.3%
Q2ai	1	0.9	87.7%	38	1.8%
Q2aii	1	0.9	88.7%	43	1.9%
Q2b	1	0.7	68.8%	40	2.4%
Q2ci	2	1.3	64.6%	59	3.2%
Q2cii	1	0.5	50.4%	43	6.4%
Q3ai	1	0.5	54.3%	43	5.9%
Q3aii	1	0.2	22.8%	26	1.7%
Q3b	2	1.2	57.8%	59	2.3%
Q3c	1	0.4	41.5%	18	4.8%
Q4	3	0.7	23.3%	59	17.2%
Q5a	1	0.6	62.6%	45	2.6%
Q5b	1	0.4	43.5%	36	2.9%
Q5ci	1	0.5	50.3%	44	4.0%
Q5cii	1	0.9	90.2%	33	1.1%
Q5di	2	0.9	43.2%	49	4.5%
Q5dii	1	0.2	17.7%	42	5.1%
Q6ai	1	0.8	75.4%	28	1.1%
Q6aii	2	0.8	38.2%	45	2.6%
Q6aiii	2	1.7	86.4%	44	1.4%
Q6bi	1	0.5	54.8%	41	6.7%
Q6bii	2	1.3	67.2%	51	3.2%
Q6c	1	0.2	21.7%	28	5.8%
Q7a	1	0.3	26.0%	23	1.4%
Q7bi	1	0.5	48.5%	38	2.6%
Q7bii	1	0.7	69.9%	34	1.5%
Q7ci	1	0.7	71.3%	50	2.9%
Q7cii	2	0.9	44.6%	45	6.5%
Q7di	2	0.3	16.2%	49	17.1%
Q7dii	1	0.4	38.2%	42	10.0%
Q8ai	1	0.9	88.1%	22	2.2%
Q8aii	1	0.8	82.5%	30	5.4%
Q8aiii	1	0.6	61.4%	51	6.0%
Q8bi	1	0.8	75.3%	35	1.5%
Q8bii	1	0.4	40.6%	37	1.8%
Q8c	2	1.4	69.1%	50	11.7%
Q9	3	0.8	26.1%	49	9.9%
Q10ai	1	0.7	68.9%	41	4.9%
Q10aii	1	0.7	71.6%	36	3.3%
Q10aiii	1	0.2	23.5%	24	3.3%
Q10bi	1	0.1	8.5%	30	12.2%
Q10bii	1	0	3.8%	16	8.6%
Q10biiiA	1	0.2	19.0%	39	7.0%
Q10biiiB	1	0.4	43.6%	43	12.3%
Q10ci	2	0.5	26.3%	53	10.2%
Q10cii	2	1.2	61.5%	39	4.7%
Q11ai	1	0.3	30.1%	39	8.3%
Q11aiiA	1	0.3	30.2%	24	11.8%
Q11aiiB	1	0.1	13.3%	21	11.8%
Q11b	1	0.3	33.3%	37	3.1%
Q11c	1	0.1	11.3%	31	7.9%
Q11d	2	0.6	31.9%	54	16.1%
Q11eiA	1	0.2	21.8%	18	3.7%
Q11eiB	1	0.2	23.8%	31	8.3%
Q11eiiA	1	0.5	50.7%	31	6.0%
Q11eiiB	1	0.3	28.2%	19	8.1%
Q11eiiiA	1	0.2	20.1%	35	15.4%
Q11eiiiB	2	1.1	57.0%	52	12.5%
Paper	85	41.5	48.8%		

2023 Advanced Higher Chemistry Section 2 Frequency of Marks Analysis

2023 Item	0	1	2	3
Q1ai	19	70	11	
Q1aii	49	51		
Q1aiiiA	18	82		
Q1aiiiB	15	85		
Q1biA	34	66		
Q1biB	31	69		
Q1biC	8	29	64	
Q1bii	18	26	57	
Q2ai	12	88		
Q2aii	11	89		
Q2b	31	69		
Q2ci	19	32	48	
Q2cii	50	50		
Q3ai	46	54		
Q3aii	77	23		
Q3b	34	17	49	
Q3c	58	42		
Q4	54	27	13	5
Q5a	37	63		
Q5b	56	44		
Q5ci	50	50		
Q5cii	10	90		
Q5di	36	42	22	
Q5dii	82	18		
Q6ai	25	75		
Q6aii	42	40	18	
Q6aiii	3	21	76	
Q6bi	45	55		
Q6bii	18	29	53	
Q6c	78	22		
Q7a	74	26		
Q7bi	52	48		
Q7bii	30	70		
Q7ci	29	71		
Q7cii	48	15	37	
Q7di	71	25	4	
Q7dii	62	38		
Q8ai	12	88		
Q8aii	17	83		
Q8aiii	39	61		
Q8bi	25	75		
Q8bii	59	41		
Q8c	20	22	58	
Q9	41	43	13	3
Q10ai	31	69		
Q10aii	28	72		
Q10aiii	76	24		
Q10bi	91	9		
Q10bii	96	4		
Q10biiiA	81	19		
Q10biiiB	56	44		
Q10ci	64	20	16	
Q10cii	23	30	46	
Q11ai	70	30		
Q11aiiA	70	30		
Q11aiiB	87	13		
Q11b	67	33		
Q11c	89	11		
Q11d	49	38	13	
Q11eiA	78	22		
Q11eiB	76	24		
Q11eiiA	49	51		
Q11eiiB	72	28		
Q11eiiiA	80	20		
Q11eiiiB	35	17	49	